Deep Learning for Physical Systems

Course Number: EN.560.617.01
Credits: 3.0
Instructor: Prof. Somdatta Goswami

Course Description

The objective of this course is to introduce the concepts of Deep Learning (DL) and scientific machine learning to the students. The course will dive into the fundamental concepts of DL and its application in solving scientific and engineering problems. Of particular interest are MLP, CNN, RNN, LSTM, Attention, Transformer, PINNs, neural operators and Graph Learning. The course will emphasize on the mathematical learning of these concepts along with applications. The course is particularly designed for senior undergraduate, masters and Ph.D. students.

Course Modules

1

- Linear algebra essentials: vectors, matrices, eigen-decompositions
- Probability and statistics for learning under uncertainty
- Numerical computation: stability, conditioning, and optimization basics

Introduction to Machine Learning

2

- Linear regression and classification
- Bias–variance trade-off and generalization
- The single-layer perceptron

Deep Feed-Forward Networks

3

- Learning the XOR problem
- Gradient-based learning and backpropagation
- Hidden units and network architecture design
- Regularization strategies
- Optimization in deep learning

Convolutional Neural Networks

4

- The convolution operator and pooling
- Variants of the basic convolution function
- Structured outputs
- The neuroscientific basis for convolutional networks

Recurrent Neural Networks

5

- Unfolding computational graphs
- Bidirectional and deep recurrent networks
- The challenge of long-term dependencies
- The Long Short-Term Memory (LSTM) architecture

Physics-Informed Neural Networks

6

- Encoding PDE residuals and boundary/initial conditions in the loss
- Training strategies and known failure modes
- Applications to forward and inverse problems in physical systems

Neural Operators

7

- From finite-dimensional networks to operator learning
- Architectures for discretization-invariant surrogates
- Generalization across geometries, resolutions, and physical regimes

Attention and Transformers

8

- The attention mechanism and self-attention
- The Transformer architecture

Graph Machine Learning *(if time permits)*

9

- Graph representations and message passing
- Graph neural network architectures
- Relevance to mesh-based and multi-body physical systems

Concluding Remarks

10

- Recap of the deep learning landscape covered in the course
- Open challenges in deep learning for physical systems
- Emerging directions and the way ahead